

Harder to Reach: A First-Person Observation of AI Polishing in Collaborative Storyboarding

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Abstract

Rough sketches preserve uncertainty, partial decisions, and traces of a maker’s thinking that help collaborators understand and respond to an emerging idea. Yet generative AI can transform these rough artifacts into polished, genre-compliant outputs before they enter collaboration. This paper examines a first-person observation from a university prototyping course, in which a teammate used generative AI to turn a rough storyboard into a polished version shortly before sharing it with our team. Through a close reading of the rough sketch, the AI-polished storyboard, and our conversation, I reflect that AI made the storyboard more legible as a deliverable while making it less available as a collaborative thinking surface. The polished version protected the maker from the vulnerability of showing in-process work, but it also obscured the hesitations, placeholders, and unresolved possibilities through which collaborators could understand and respond to their thinking. This reflection brings up design questions for creativity support tools around when to intervene, what process traces to preserve, and how to make rough work shareable without forcing exposure.

Keywords

sketching, generative AI, reflection-in-action, collaborative design, design education

1 Introduction

Before one of our team meetings, my teammate sent a short message in our group chat: “sketch + AI.” Ten minutes later, a rough hand-drawn storyboard had become a polished narrative — easier to read, harder to think with. This paper is about that ten-minute interval and what it did to our collaboration. The polished storyboard arrived looking like a finished account of a user scenario. Our meeting was for brainstorming, however, and what we needed to see was where my teammate was still uncertain, so we could help each other think. The AI had made the artifact more legible while making the person who made it harder to reach.

The unfinishedness of sketches is valuable. In Schön’s account of reflective practice, a sketch records how the designer thinks [10]. The sketch “talks back,” surprising its maker, redirecting attention, opening possibilities that were not there before the pencil moved. Goldschmidt [7] shows this in practice: designers oscillate between seeing what they drew as something and reasoning about it, a back-and-forth that only works when the sketch stays open. As Buxton [1] puts it, a sketch derives its value from ambiguity and disposability rather than resolution.

This reflective function is also important in collaboration. A rough sketch enters a room carrying undecided things in a form

that the whole team can interrogate. One teammate reads a scenario as a problem of information overload; another sees it as a problem of trust. That divergence is information about the design space. Suchman’s concept of plans as resources rather than programs [13] applies here: rough sketches provide scaffolding for situated, emergent reasoning. Goel [6] draws a relevant distinction that lateral transformations reinterpret a sketch to generate new ideas, while vertical transformations refine it toward a fixed goal. Early collaborative design depends on keeping the artifact open to lateral movement.

Generative AI compresses the distance between a rough sketch and a resolved one. A rough drawing can become a polished storyboard in seconds. What gets erased in that transformation is precisely what made the sketch useful for the team to think through together: its ambiguity, openness, and capacity to carry unresolved questions. Lin et al. [8] note that AI-generated images often appear too complete to iterate upon, narrowing the design space designers feel they can still move within. Ford and Bryan-Kinns [4] identify reflection on current process and reflection through experimentation as central dimensions of creative experience; both depend on the artifact remaining available in its uncertain state. Ford et al. [5] further note reflective thought is often tacit and hard to express, which is what the rough sketch helps make visible, and what polishing erases.

In this paper, I reflect on this moment to ask what happens when AI intervenes at the threshold between private making and public sharing, and how polish can protect a maker from the vulnerability of showing the work while also obscuring the traces of hesitation, uncertainty, and partial decision-making that enable collaborative reflection.

2 A First-Person Observation

I am a student at the University of Wisconsin–Madison, majoring in Data Science and Information Science. This semester, I took a product prototyping course in which small teams designed a digital product. One step was building storyboards to understand our users. Our team of three held self-organized project meetings on Zoom.

2.1 Before the Screen Share

Before one of our sessions, my teammate—whom I will refer to as Momo, a pseudonym used with their consent—sent a message in our group chat: “sketch + AI.” They explained that their drawing skills were limited to stick figures and asked for ten minutes to polish their sketch before sharing. I told them the sketch was fine to use directly; this was not a space where anyone would judge artistic skill. They replied that it was too rough and asked for the time anyway.

2.2 From Rough Sketch to AI-polished Storyboard

Ten minutes later, they shared their screen. The image was a clean four-panel storyboard with detailed characters, clear facial expressions, a polished interface, captions under each panel, and a readable emotional arc from anxiety to relief (Fig. 1). It identified a pain point, introduced an action, presented a design opportunity, and ended with a successful outcome. Looking at it as a teammate, I found it difficult to know how to respond. I could evaluate the storyboard as a finished explanation, but I could not easily see what Momo wanted us to help think through. I did not know which parts were Momo's decisions, which were AI inferences, and which were simply the conventions of what a storyboard is supposed to look like. At that moment, I was also confused by what made the AI-polished version feel more acceptable to Momo than their original sketch. I asked them: Maybe just use your original draft instead? They hesitated, but shared it. When the rough sketch appeared alongside the AI version, Momo described the two images as "two completely different species" and sent a string of frustrated emoji.

2.3 What the Rough Sketch Made Visible

The rough sketch contained real work. It held several scene fragments: a clock, a laptop, Canvas, a possible "Study Hub all-in-one," word placeholders, a user sitting or lying back, and a folder placed on a table. These marks suggested a design space without fully specifying the story—the central problem could have been missing announcements, switching between platforms, hidden deadlines, or the emotional labor of checking multiple systems. The sketch left those possibilities open. The AI-polished version closed them by producing a coherent problem-solution narrative in which the user begins anxious, remembers the dashboard, discovers an urgent reading assignment, and completes it on time. I told Momo that the AI version was very neat, though I could not quite see their thinking process in it. They laughed, then said more quietly that what I said made a lot of sense.

2.4 Reflection Through Comparison

Momo's rough sketch was still a private working draft: incomplete, tentative, and full of unresolved possibilities. The AI-generated version appeared just as that rough draft was about to enter our conversation. It changed how the storyboard looked, and it changed what I felt able to respond to as a collaborator. At the same time, I should acknowledge that the polished storyboard did not stop conversation entirely; it led to the exchange I am describing. When I asked Momo to share the rough sketch, and when Momo recognized what had changed between the two versions, we had a small moment of collaborative reflection. That reflection happened because we stepped back from the polished version, looked at the rough sketch, and compared the two side by side. Without that comparison, Momo's thinking process would have remained much harder to see. And in a team with less trust, less shared context, or more pressure to accept polished work as finished work, that conversation might never have happened.

3 What AI Changes

3.1 From Fragments to Resolved Narrative

Momo's rough sketch made several possible stories visible without committing to one. The fragments suggested a design space, though they did not determine what the central problem was or how the scenario should unfold. The AI-polished version made one interpretation appear coherent, organizing the panels into a recognizable storyboard structure. This made the storyboard more legible, but it also made the underlying commitments harder to discuss. It became unclear which parts reflected Momo's decisions, which came from the AI's inferential filling-in, and which were inherited from the genre conventions of storyboarding. Wadinambiarachchi et al. [14] report a related effect in controlled studies: exposure to AI-generated images during ideation increased design fixation and reduced the variety and originality of subsequent ideas, with the focus of fixation shifting onto the AI's outputs. Polishing is not inherently the problem. If Momo had redrawn the sketch by hand, the cleanup might itself have been reflective—deciding which scenes mattered, what sequence to use, what emotions to name, what outcome to show. The AI performed that interpretive work instead. The artifact became polished, and the decisions through which a rough sketch becomes a storyboard were no longer visible to collaborators.

3.2 The Person Behind the Artifact

AI polish also changed my relationship to Momo as the reader of the artifact. The polished storyboard was more conventionally recognizable as a storyboard. In that sense, it communicated more successfully. It also made the traces of its making less visible, and those traces are one of the ways collaborators sense the person behind the artifact. In the rough sketch, the unfinished marks, placeholders, and uneven levels of detail gave me a picture of Momo's thinking. They showed where they had paused, what they had not yet decided, and where the team might enter the work. The rough sketch represented a person trying to think through a user scenario. The placeholders and unresolved scenes were a record of someone in the middle of a problem, an intermediate artifact that directly encodes the first-hand experience of making. Sharing it would have been a small act of vulnerability, a sense of willingness to be seen not-knowing.

The loss felt relational. A rough sketch says, implicitly: here is where I am, help me think from here. The polished storyboard said something closer to: here is a finished account, evaluate whether it works. That difference changed what I felt able to do as a collaborator. The polish that protected Momo from exposure also made them harder to reach.

3.3 Timing and Social Pressure

The timing of the AI intervention matters because the value of polish depends on what the artifact is supposed to do. If the goal is to prepare a class deliverable, then polishing may be reasonable. In that context, Momo's decision to use AI responded to a real expectation that storyboards should communicate clearly, as well as to a real vulnerability around sharing rough work in a setting where artifacts can become graded.

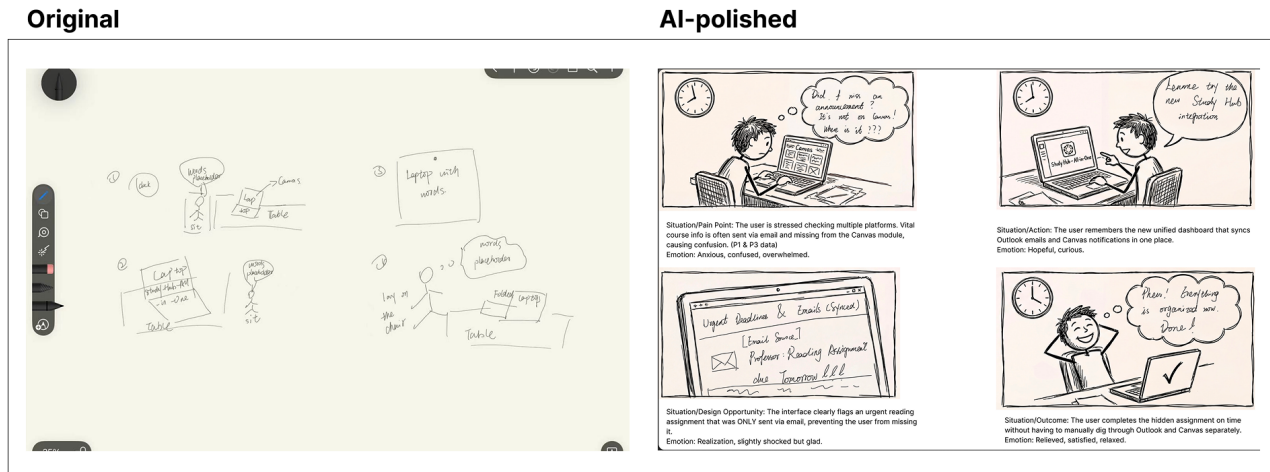


Figure 1: The original sketch and the AI-polished storyboard. The AI version is more legible as a storyboard deliverable, but it stabilizes the scenario before the team has used the sketch as a thinking surface.

In our case, however, the intervention happened before the team had clarified what kind of work the storyboard needed to do. Momo reached for AI in the ten minutes before sharing, at the moment when the sketch was about to move from private thinking into collaborative attention. The tool made the artifact socially safer to show, but it did so at the stage when its incompleteness might have been most useful for the team. Before we could ask what the sketch was trying to explore, AI had already helped turn it into something that looked more like a finished account.

External feedback shapes the artifact and the creator’s relationship to their own judgment. E et al. have shown that even well-timed computational feedback in design can displace self-initiated reflection, leading novices to rely on external evaluation rather than developing their own judgment [2]. For Momo, AI polish functioned less as a cleanup tool than as a way to manage the uncertainty of showing the sketch, a confidence-supporting move. It made the artifact feel safer to present, while also reducing the visibility of the thinking traces that would have allowed collaborators to respond to Momo’s process rather than only to the polished output.

According to Momo, their prior environments had rewarded finished things such as submitted assignments. AI did not create that evaluative pressure; it provided a fast, low-effort way to manage it. Palani and Ramos [9] observe that creative practitioners’ roles and workflows are shifting as GenAI moves into earlier stages of their process, which raises new questions about when intermediate artifacts count as work-in-progress and when they begin to function as deliverables. Without AI, Momo might have handled the same anxiety differently, perhaps by spending longer cleaning up the sketch by hand or by sharing less of their thinking. If the problem is fundamentally about social-evaluative pressure in design education, then traceability features in AI tools, while potentially valuable, will not solve it on their own. The challenge is partly about what tools afford, and partly about what collaborative environments make safe.

4 Reflexive Positioning

My reading of what had been lost was shaped by my own relationship to sketching. As a visual thinker, I often use rough sketches as sensemaking tools. When discussing research ideas with my advisor, I draw small illustrations to communicate what I mean, even when the drawings are messy. I have learned to treat roughness as part of thinking. Momo, on the other hand, had learned something different. Based on our later conversation, their prior environments had often treated sharing as a form of submission for evaluation. Showing a sketch meant being seen—and being judged.

I also need to acknowledge what my question—“maybe just use your original draft?”—did to our dynamic. I asked because the polished storyboard had made their thinking unclear to me. Asking also placed Momo in a position of accounting for their process in a way I was not accounting for mine. Their rough sketch was a spontaneous exposure; this paper is a retrospection. That asymmetry matters for how I interpret what I saw.

Writing this paper has itself been a reflective act. At the moment, I experienced the polished storyboard as a loss; something had been taken away from our collaboration. The act of writing forced me to separate what I *observed*, which is a shift in the artifact’s form and my own difficulty responding to it, from what I *inferred*, which is that Momo’s reflective process had been foreclosed. I do not know what Momo experienced during those ten minutes with the large language model. They may have reflected extensively while prompting and selecting the output. My account therefore contributes a situated account of how AI polish can be experienced by a collaborator, rather than a complete account of the maker’s creative process.

I should also note that Momo has read this paper and approved my account, including the figures. The reading is still mine, though, not a joint analysis.

5 Open Questions

This observation brings up questions for researchers and tool designers who work on creativity support in contexts where AI mediates the transition between thinking and sharing.

When does a storyboard need to be a thinking surface, and when does it need to be a deliverable? In Momo’s case, the same artifact was caught between two expectations. Future work might attend to transitions between artifact roles: when do designers treat sketches as private thinking tools, collaborative surfaces, or presentational outputs? How might AI systems support that movement without collapsing the gradient?

What traces should remain available when AI helps formalize an artifact? If AI helps a designer polish a storyboard, what should remain visible from the earlier process? Should collaborators be able to see placeholders, alternative scene orders, rejected captions, uncertainty marks, or annotations about what the designer wants feedback on? Collaboration depends on what an artifact shows, and also on what kinds of thinking it makes reachable. In creative practice, low-fidelity records are often kept incomplete on purpose—practitioners leave information out because the gaps create space for productive variation and adaptation [12].

What if AI scaffolded reflection rather than generating polish? Instead of replacing Momo’s sketch with a polished artifact, AI might have helped them see that the original was already clear enough to share, named what was legible in it, or marked places where multiple interpretations were still possible. Rather than producing a finished storyboard, a system might say: “This scene seems important but still open,” or “You might ask your team whether the emotional arc should begin here or later.” In Fleck and Fitzpatrick’s terms [3], such a tool would aim to support *dialogic* or *transformative* reflection—helping the designer examine their own assumptions and consider alternatives—rather than merely enabling *revisiting*, where the artifact is reviewed without deeper questioning. This alternative carries its own risks: the model would still be making interpretive choices about which traces matter, which is itself a subtler form of stabilization.

How can AI make early work feel safe without erasing its traces? Showing early work can feel risky, especially for students whose confidence has been shaped by evaluative environments. A tool that simply says “show your rough work” may reproduce the vulnerability that made polish attractive in the first place. Recent work on happiness in creativity support tools reminds us that creative technologies should be evaluated by output quality and efficiency, as well as by how they shape joy, satisfaction, and negative feelings in creative practice [11]. In Momo’s case, AI polish may have reduced the discomfort of sharing a rough sketch, while it also weakened the collaborative visibility of the maker’s process. Since sharing rough work can build rapport among collaborators, one possibility is to maintain visible connections between polished outputs and the rough decisions that preceded them—showing layers, alternatives, or uncertain regions—so that sharing becomes a structured part of the tool, less of a personal risk the designer must take alone.

Acknowledgments

I greatly appreciate my teammates, especially Momo, for generously allowing me to reflect on this moment, for sharing both the polished storyboard and the rough sketch behind it, and for the conversations that sparked this reflection.

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